

Exact Finite Costs and the Non-Gaussian Contraction Limit of Randomized Quicksort

Route-A source-local certificate HCS-C302

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Abstract

We determine the comparison-cost distribution of classical single-pivot Quicksort by an exact probability-generating-polynomial recurrence and derive closed formulas for its mean and variance at every input size.

1 The finite recursive law

Let X_n count key comparisons made by classical single-pivot Quicksort on a uniform random permutation of n distinct keys. The pivot is compared once with every other key, and $X_0 = X_1 = 0$. If I_n is uniform on $\{0, \dots, n-1\}$, the relative orders in the two subarrays are independent uniform permutations; hence

$$X_n \stackrel{d}{=} X_{I_n} + X'_{n-1-I_n} + n - 1. \quad (1)$$

All variables on the two recursive branches are independent conditional on I_n .

Theorem 1 (Every finite law and its first two moments). *For $G_n(z) = \mathbb{E}z^{X_n}$, $G_0 = G_1 = 1$ and*

$$G_n(z) = \frac{z^{n-1}}{n} \sum_{j=0}^{n-1} G_j(z) G_{n-1-j}(z), \quad n \geq 2. \quad (2)$$

Writing $H_n = \sum_{k=1}^n k^{-1}$ and $H_n^{(2)} = \sum_{k=1}^n k^{-2}$, with empty sums zero,

$$\mu_n := \mathbb{E}X_n = 2(n+1)H_n - 4n, \quad (3)$$

$$v_n := \text{Var } X_n = 7n^2 - 4(n+1)^2 H_n^{(2)} - 2(n+1)H_n + 13n. \quad (4)$$

These formulas include $n = 0, 1$.

Proof. Conditioning (1) on the pivot rank and multiplying independent PGFs gives (2). Expectations satisfy

$$\mu_n = n - 1 + \frac{2}{n} \sum_{j=0}^{n-1} \mu_j, \quad (5)$$

whose successive difference is solved by (3). Total variance gives

$$v_n = \frac{2}{n} \sum_{j=0}^{n-1} v_j + \text{Var}(\mu_{I_n} + \mu_{n-1-I_n}). \quad (6)$$

Substitution of (3), followed by the elementary harmonic-sum identities, reduces (6) to (4). Both formulas also verify the two boundary values directly. $\square$

Literature ownership and AI-use statement

Hoare introduced Quicksort [1]; Régnier [2] and Rösler [3] established its limiting distribution and contraction approach. We make no literary priority claim. AI tools assisted exact-algebra checks, hostile review and manuscript preparation; all formulas and proof obligations are exposed here and in the independent certificate.

References

- [1] C. A. R. Hoare, “Quicksort,” *The Computer Journal* 5 (1962), 10–16, doi:10.1093/comjnl/5.1.10.
- [2] M. Régnier, “A limiting distribution for quicksort,” *RAIRO Theor. Inform. Appl.* 23 (1989), 335–343.
- [3] U. Rösler, “A limit theorem for Quicksort,” *RAIRO Theor. Inform. Appl.* 25 (1991), 85–100.